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Original article

Prevalence and factors associated with prediabetes and diabetes mellitus among adults: Baseline findings of PURE Malaysia cohort study

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ABSTRACT

Background: Diabetes mellitus (DM) is a silent killer that is responsible for almost 1.6 million deaths annually, particularly among those who are undiagnosed during its early stage. Prediabetes has become a growing public health concern due to its potential to progress to DM. This study thus aimed to determine the prevalence of prediabetes and DM and their associated factors among Malaysian adults.

Methods: A cross-sectional study was conducted among adults 35–70 years of age residing in rural and urban areas in Malaysia. Blood samples (finger prick test) and physical examinations were conducted on 4982 participants who consented to participate in this study. A pre-validated questionnaire consisting of the International Physical Activity Questionnaire and medical history was used to assess physical activity level and family history of DM, respectively. Multinomial logistic regression models were used to identify the factors associated with prediabetes and DM.

Results: The prevalence of prediabetes and DM were 10.8% and 11.9%, respectively. Participants who were ≥50 years old, male, Malay, or physically inactive or had hypertension or a family history of DM had higher odds of having prediabetes and DM. Unique to DM, individuals with a lower educational level were more likely to have DM.

Conclusions: Prediabetes health screening is critical in the Malaysian population. Early detection of prediabetes promotes early intervention, including lifestyle modifications, to prevent progression to DM. The factors associated with prediabetes and DM identified in this study will assist in disease prevention and facilitate more efficient management strategies in this population.

1. Introduction

Diabetes mellitus (DM) is a condition characterized by elevated levels of blood glucose that may lead to serious damage to the heart, blood vessels, eyes, kidneys and nerves over time.[1] According to the World Health Organization (WHO), approximately 422 million people worldwide have DM, which caused 1.6 million deaths annually. The global prevalence of DM among adults aged 18 and older has risen dramatically, reaching 8.8% in 2017 from 4.7% in 1980.[1] This number is expected to further increase to 9.9% in 2045, affecting more than 600 million people worldwide.[2] Studies have shown that factors ranging from genetics to lifestyle are associated with DM and include obesity, age, family history, hypertension, perceived stress, smoking, poor diet and physical inactivity.[3–5]

In Malaysia, the national data on DM as reported in the National Health and Morbidity Survey (NHMS) have shown an increasing trend for nearly the past two decades with a prevalence that more than doubled between 1996 and 2015.[6] The most recent NHMS revealed that

Abbreviations: DM, Diabetes mellitus; PURE, Prospective Urban Rural Epidemiology; NHMS, National health and morbidity survey; IPAQ, International Physical Activity Questionnaire; FBS, fasting blood sugar; RBS, random blood sugar.

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one in every five adults reported having DM and its prevalence rose from 13.4% in 2015 to 18.3% in 2019.[7,8] DM substantially increases the risk of cardiovascular diseases, a known leading cause of death in Malaysia accounting for 35% of all Malaysian mortalities.[9]

The prevalence, risk factors, and economic and social impacts of DM have been studied extensively; however, the increasing trends of DM continue to cause concern among global and local health practitioners. Filling the gaps in the data on the magnitude and local trends of prediabetes may provide a critical understanding of how to enhance local DM prevention and disease management strategies. This study aimed to address these gaps. Thus, the objective of this study was to assess the prevalence and factors associated with prediabetes and DM among adults aged 35–70 years residing in rural and urban areas of Malaysia.

2. Methodology

2.1. Study design and population

This was a sub-study under the Prospective Urban Rural Epidemiology (PURE) study that involves 21 countries, including Malaysia, and aims to determine the impact of societal influences on the prevalence of select non-communicable diseases. The comprehensive methodology of the overall PURE study has been explained in detail in previous studies.[10,11] This community-based study included adults aged 35–70 years.

2.2. Patient and public involvement

The respondents were recruited from select urban and rural areas with the assistance of local community leaders. Health screening and health promotion booths were set up in the communities' assembly halls, where interested and eligible participants were provided information on the study. After they agreed to participate and provided written consent, participants' medical history was obtained and a basic physical examination was conducted. All data were obtained through face-to-face interview sessions by well-trained research assistants using a standardised and verified set of questionnaires.

2.3. Study tool

The questionnaires consisted of two sections, the a) Adult Questionnaire and b) International Physical Activity Questionnaire (IPAQ). The Adult Questionnaire was developed by the Population Health Research Institute (PHRI) and revised by the Malaysian team of researchers to obtain information related to the socio-demographic characteristics of the respondents. Information related to family history of hypertension was also recorded. The IPAQ was used to collect data on the respondents' metabolic rates (METs; min/week), and respondents' physical activity level was further categorised as inactive or active.

2.4. Study procedure

Blood glucose level was measured by a trained research assistant using a calibrated GlucoSure Plus blood glucose monitor (Apex Biotechnology Corporation, Taiwan). The measurement was performed and recorded as fasting blood sugar (FBS) if the participants fasted for at least 10 h prior to the test and random blood sugar (RBS) if the participants did not fast. Participants who reported having a previous diagnosis of DM with verification of treatment and medication prescription were considered to be in the DM group. Those who had newly detected high blood glucose from the finger prick test and those whose blood glucose ranged from 5.6 to <7.0 mmol/L (FBG) and 7.8 to < 11 mmol/L (RBS) were in the prediabetes group.

2.5. Statistical analysis

The data were analysed using SPSS version 26 (IBM, Armonk, NY, USA). The general characteristics of participants were descriptively analysed and presented as numbers (and corresponding percentages). A multinomial logistic regression analysis was performed to investigate the potential determinants of prediabetes and DMs. Adjusted odds ratios (AORs) and 95% confidence intervals (CIs) were calculated. The statistical significance level was set at $p < 0.05$. The model was adjusted for age, gender, ethnicity, education level, physical activity, smoking status, alcohol intake, co-morbidities of hypertension and a family history of DM.

3. Results

Of the 4982 adults who participated in the study, a total of 537 (10.8%) and 591 (11.9%) reported having prediabetes and DM, respectively, as shown in Table 1. Similar demographic characteristics were seen in the DM and non-DM groups, with the majority of both groups being female, Malay and currently married. The exception to the similar demographics was education level, as the DM group had more participants with higher education levels than participants with lower education levels.

Table 2 presents the factors associated with prediabetes and DM. For prediabetes, respondents 50 years of age and older had nearly two times the odds (adjusted odds ratio (AOR): 1.717; 95% confidence interval (CI): 1.381–2.133) of having prediabetes compared to those under 50 years of age. Malay respondents were three times more likely (AOR: 3.190; 95% CI: 2.14–4.755) to have prediabetes than respondents from other ethnic groups. Males had higher odds (AOR: 1.542; 95% CI: 1.237–1.922) of having prediabetes than females. Those who reported being physically inactive were more likely (AOR: 1.249; 95% CI: 1.033–1.511) to have prediabetes than those who were physically active. Respondents who reported having hypertension as a comorbidity

Table 1
General characteristics of participants (N = 4982).

Characteristics	Total	Frequency (%)		
		Non-DM	Pre-DM	DM
N (%)	4982	3854 (77.4)	537 (10.8)	591 (11.9)
Age				
≥50	2943 (59.1)	2144 (72.9)	380 (12.9)	419 (14.2)
<50	2039 (40.9)	1710 (83.9)	157 (7.7)	172 (8.4)
Gender				
Male	2151 (43.2)	1622 (75.4)	266 (12.4)	263 (12.2)
Female	2831 (56.8)	2232 (78.8)	271 (9.6)	328 (11.6)
Marital status				
No	491 (9.9)	356 (72.5)	59 (12)	76 (15.5)
Yes	4491 (90.1)	3498 (77.9)	478 (10.6)	515 (11.5)
Ethnicity				
Malay	4246 (85.2)	3181 (74.9)	508 (12)	557 (13.1)
Non-Malay	736 (14.8)	673 (91.4)	29 (3.9)	34 (4.6)
Education level				
Low	2432 (48.8)	1854 (76.2)	264 (10.9)	314 (12.9)
High	2550 (51.2)	2000 (78.4)	273 (10.7)	277 (10.9)
IPAQ				
Inactive	1884 (37.8)	1428 (75.8)	221 (11.7)	235 (12.5)
Active	3098 (62.2)	2426 (78.3)	316 (10.2)	356 (11.5)
Smoking				
No	3837 (77)	2973 (77.5)	412 (10.7)	452 (11.8)
Yes	1145 (23)	881 (76.9)	125 (10.9)	139 (12.1)
Alcohol				
No	4761 (95.6)	3659 (76.9)	527 (11.1)	575 (12.1)
Yes	221 (4.4)	195 (88.2)	10 (4.5)	16 (7.2)
Hypertension				
Yes	1291 (25.9)	833 (64.5)	196 (15.2)	262 (20.3)
No	3691 (74.1)	3021 (81.8)	341 (9.2)	329 (8.9)
Family history of DM				
Yes	1033 (20.7)	706 (68.3)	117 (11.3)	210 (20.3)
No	3949 (79.3)	3148 (79.7)	420 (10.6)	381 (9.6)

Table 2
Multinomial logistic regression analysis of factors associated with prediabetes and DM (N = 4982).

Variables	Prediabetes			DM		
	β	AOR (95% CI)	p-value	β	AOR (95% CI)	p-value
Age						
≥50	0.54	**1.717** (1.381–2.133)	**< 0.001**	0.510	**1.666** (1.346–2.062)	**< 0.001**
<50						
Gender						
Male	0.433	**1.542** (1.237–1.922)	**< 0.001**	0.260	**1.297** (1.039–1.619)	**0.022**
Female						
Marital status						
No	0.224	1.251 (0.921–1.701)	0.152	0.310	**1.363** (1.025–1.812)	**0.033**
Yes						
Ethnicity						
Malay	1.16	**3.190** (2.14–4.755)	**< 0.001**	1.088	**2.967** (2.03–4.338)	**< 0.001**
Non-Malay						
Education level						
Low	−0.08	0.923 (0.753–1.131)	0.439	0.182	1.199 (0.981–1.467)	0.077
High						
IPAQ						
Inactive	0.223	**1.249** (1.033–1.511)	**0.022**	0.215	**1.239** (1.028–1.494)	**0.025**
Active						
Smoking						
No	0.244	1.276 (0.992–1.641)	0.058	0.099	1.104 (0.858–1.419)	0.442
Yes						
Alcohol						
No	0.52	1.681 (0.858–3.294)	0.130	0.086	1.09 (0.621–1.912)	0.764
Yes						
Hypertension						
Yes	0.61	**1.841** (1.51–2.244)	**< 0.001**	0.905	**2.473** (2.05–2.983)	**< 0.001**
No						
Family history of DM						
Yes	0.296	**1.345** (1.066–1.696)	**0.012**	1.035	**2.816** (2.298–3.449)	**< 0.001**
No						

$R^2 = 11.1\%$.

were nearly two times more likely (AOR: 1.841; 95% CI: 1.51–2.244) to have prediabetes than respondents without hypertension. Those with a family history of DM were more likely (AOR: 1.345; 95% CI: 1.066–1.696) to have prediabetes than those without a family history of DM.

For DM, respondents who were 50 years of age and older had nearly two times the odds of having DM (AOR: 1.666; 95% CI: 1.346–2.062) compared to those under 50 years of age. Malays were nearly three times more likely (AOR: 2.967; 95% CI: 2.03–4.338) to report having DM than non-Malays group. Compared to females, males were more likely to have DM (AOR: 1.297; 95% CI: 1.039–1.619). Unmarried respondents had higher odds (AOR: 1.363; 95% CI: 1.025–1.812) of having DM than respondents who were married. Physically inactive individuals were more likely to report having DM (AOR: 1.239; 95% CI: 1.028–1.494) than those who were physically active. Respondents with hypertension had more than two times the odds (AOR: 2.473; 95% CI: 2.05–2.983) of having DM as compared to those without hypertension. Those who had a family history of DM were nearly three times more likely to have DM (AOR: 2.816; 95% CI: 2.298–3.449) than those without family history of DM.

4. Discussion

This study revealed a prevalence of prediabetes and DM in the study population of 10.8% and 11.9%, respectively. The prevalence of DM was slightly lower than the two most recent NHMS reports, which stated a DM prevalence of 18.3% in 2019 and 13.4% in 2015.[7,8] This result was expected, as the NHMS encompassed a much wider age population compared to the present study, which included only adults between 35 and 70 years of age. A smaller study of adults in fishing communities in Penang, Malaysia found a higher DM prevalence of 19.6%, while the prediabetes prevalence of 10.1% was lower than the prevalence reported in this study.[9]

Older individuals (aged 50 years and above) were shown to have a higher risk of having both prediabetes and DM. This finding is supported by previous studies and NHMS reports that found that the prevalence of DM was 11.8% in those 30–49 years of age compared to 39.9% in those above 50 years of age.[12,13] This marked difference has been suggested to be due to reduced insulin sensitivity and impaired pancreatic islet function with ageing.[14] This effect of ageing has been demonstrated by the role of factors such as arginine vasopressin or its C-terminal fragment, called copeptin, in affecting the sensitivity of insulin produced in older people.[15] In addition, limited ability to be physically active and an unbalanced diet high in carbohydrates and sugars in the older population may contribute to central obesity, which is responsible for insulin resistance, a direct associated factor of DM.

Males exhibited a higher risk of having prediabetes and DM compared to women. This finding is consistent with various studies that showed a link between testosterone hormone and development of DM in men, with lower testosterone levels leading to greater risk of DM.[16] It is well established that testosterone deficiency promotes the development of visceral obesity, which causes insulin resistance and metabolic syndrome in men.[17,18] Moreover, uncontrolled dietary intake of sugary foods, lack of physical activity and smoking may increase the likelihood of men having prediabetes and DM compared to women.

This study revealed that Malays had three times higher odds of having prediabetes and DM compared to non-Malays. Although this finding contradicts the NHMS report, the result was expected, as the respondents involved in this study were majority from the Malay ethnicity. According to the NHMS, the highest prevalence of DM was among Indians (18.5%), followed by Malays (11.0%), Chinese (8.5%) and Bumiputera Sarawak (7.9%).[8]

Being physically active, encompassing any type of exercise and intensity, has been shown to reduce the onset of DM by 25%–40%.[19–22] This may be explained by several biological mechanisms that improve energy balance and reduce adiposity, the primary risk factor for DM. Physical activity increases skeletal muscle glucose uptake by translocation of GLUT4 glucose transporters to the skeletal muscle cell

membranes, increasing the activity of glycogen synthase and contributing to reduced blood glucose levels.[23,24] Furthermore, physical activity increases the secretion of interleukin (IL)-6 from muscle cells, which have anti-inflammatory effects through the inhibition of tumor necrosis factor (TNF)-a and IL-1b, and reduces TNF-induced insulin resistance, improving the regulation of glucose homeostasis.[25,26]

Individuals with comorbidities such as hypertension were more likely to have prediabetes and DM. Studies have suggested that hypertension is twice as frequent in patients with DM compared to those without DM.[4,27–29] Hypertension has also been associated with cardiovascular diseases and is emerging as a risk factor for DM. A simple explanation for this association is the inefficiency of insulin production in patients with DM that leads to glucose accumulation in the bloodstream. This eventually causes damage to the kidneys and blood vessels, which are responsible for maintaining healthy blood pressure. In addition, individuals with hypertension may experience symptoms such as headaches, dizziness, shortness of breath or chest pain, making them less likely to be physically active and thus prone to obesity, a known cause of DM.

Respondents with a family history of DM were more likely to have prediabetes and DM compared to those without a family history. Studies performed across multiple study designs and ethnic groups have found a twofold to sixfold increased risk of DM associated with a positive family history.[3,30,31] The genetic factor causing this association is still unclear. Another explanation could be related to lifestyle, as families tend to share similar habits and lifestyles, which include dietary intake and physical activity.[32] For instance, a family that practises a healthy and balanced diet and is physically active may raise children with less risk of obesity and thus a reduced risk of DM. Families with higher levels of awareness, higher health literacy and healthier lifestyles that include controlling the intake of carbohydrate-rich foods and sugary drinks surely help to reduce the risk of family members developing prediabetes and DM.

The main limitation in this study was exclusion of BMI as one of the predictors of pre-diabetes and diabetes because BMI has limited clinical value for accurately defining obesity, especially in the Asian population. Obesity is associated with increased body growth due to mismatching overfeeding and physical activity as well as increased lipid deposition in subcutaneous and visceral adipose depots.[33] However, the detrimental effects of long-term obesity influence adult diabetes are not entirely clear to people.[34] Therefore, it limits the researcher's capacity to understand the relationship between weight and health.[35] This is because BMI does not measure the degree of fat storage, which is more significantly related to health risk.[36] The Quetelet Index aka BMI less than 18.5 is underweight, 18.5 to 24.9 is a normal weight, 25 to 29.9 is overweight, and a score of 30 and above is considered obese receives plenty of criticism. It is less applicable to very tall or short, muscular people and Asian ethnicity such as this study population that have an average height of 156.4 cm. It is interesting to note that Asian individuals are susceptible to diabetes at lower BMIs than White individuals.[37]

5. Conclusion

Despite the various action plans formulated to reduce DM in the community, data indicate a continuous rise in the prevalence of DM and prediabetes in Malaysia over the past two decades. A proactive approach is needed to raise awareness of the importance of early detection of prediabetes, particularly among individuals with risk factors such as a family history of DM. Early detection enables health promotion and early interventions such as lifestyle modifications, which reduce the risk of progression to DM. As for patients with existing DM, lifestyle changes and advice on medication adherence are needed to minimise the complications of DM.

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Ethical approval

The Hamilton Health Sciences Research Ethics Board approved the study protocol (PHRI; grant no. 101414), and local ethics clearance was obtained from the Research and Ethics Committee of Universiti Kebangsaan Malaysia (UKM) Medical Center (project code: PHUM-2012-01) and the Research Ethics Committee of Universiti Teknologi Mara (UiTM).

CRediT authorship contribution statement

Rosnah Ismail: Conceptualization, Methodology, Resources, Funding acquisition, Writing – original draft, Writing – review & editing, All authors have read and agreed on the final version of the manuscript. **Noor Hassim Ismail:** Conceptualization, Methodology, Funding acquisition, All authors have read and agreed on the final version of the manuscript. **Azmi Mohd Tamil:** Conceptualization, Methodology, Data collection, Writing – review & editing, All authors have read and agreed on the final version of the manuscript. **Mohd Hasni Jaafar:** Conceptualization, Methodology, Data collection, Funding acquisition, Writing – review & editing, All authors have read and agreed on the final version of the manuscript. **Zaleha Md Isa:** Conceptualization, Methodology, Writing – review & editing, All authors have read and agreed on the final version of the manuscript. **Nafiza Mat Nasir:** Methodology, Resources, Data collection, All authors have read and agreed on the final version of the manuscript. **Farnaza Ariffin:** Writing – review & editing, All authors have read and agreed on the final version of the manuscript. **Anis Safura Ramli:** Writing – review & editing, All authors have read and agreed on the final version of the manuscript. **Najihah Zainol Abidin:** Formal analysis, Writing – original draft, Writing – review & editing, All authors have read and agreed on the final version of the manuscript. **Nurul Hafiza Ab Razak:** Formal analysis, Writing – review & editing, All authors have read and agreed on the final version of the manuscript. **Khairul Hazdi Yusof:** Resources, Data collection, Formal analysis, All authors have read and agreed on the final version of the manuscript.

Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as potential conflicts of interest.

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